

Which chemical properties below are those of the alkaline-earth metals?

- I. Form ions with a +1 charge
- II. Readily oxidized to an oxidation state of +2
- III. Increasing reactivity with increasing atomic number
- IV. Decreasing reactivity with increasing atomic number
- V. Soft Group 1 metals that form basic hydroxides in water
- VI. Group 2 metals that form hydrogen gas when reacting with water

☐ A. I, III, and V only [6%]

☐ B. II, IV, and VI only [29%]

☐ C. I, IV, and V only [5%]

✓ ☒ D. II, III, and VI only [57%]

Correct

58% answered correctly

Explanation:

Selected chemical properties of alkaline-earth metals

Group 1		Group 2
H		Be
Li		Mg
Na		Ca
K		Sr
Rb		Ba
Cs		

Increasing reactivity ↓

Ion formation (oxidation)
 $M \rightarrow M^{2+} + 2e^{-}$

Reactions with water*
 $M(s) + H_2O(g) \rightarrow MO(s) + H_2(g)$
 $M(s) + 2H_2O(l) \rightarrow M(OH)_2(aq) + H_2(g)$

*Be participates only in Reaction 1

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The **alkaline-earth metals** are a **highly reactive** family of metals occupying **Group 2** of the periodic table. Because the alkaline-earth metals have a low **electronegativity** and possess only **two valence electrons**, these metals have a low first and second **ionization energy**. As a result, the alkaline-earth metals readily lose both valence electrons (oxidize), which results in an **oxidation state** of +2 (**Number II**).

As the atoms in the alkaline-earth metal group **increase in size** moving down the column, the valence electrons occupy an increasingly **higher energy level** and therefore are less tightly bound. Accordingly, less energy is required to remove the electrons from the valence shell of larger alkaline-earth metals. Consequently, the **reactivity** of the alkaline-earth metals **increases** with **increasing atomic number** (**Number III**).

When reacting with water, the alkaline-earth metals of Group 2 form either a **metal oxide** or a very basic metal **hydroxide solution** while **releasing hydrogen gas** in the process (**Number VI**).

(Number I) Forming ions with a +1 charge is a property of the **alkali metals** (Group 1) rather than the alkaline-earth metals (Group 2).

(Number IV) Both alkali metals and alkaline-earth metals show an *increase*, not a decrease, in reactivity with increasing atomic number.

(Number V) Although both alkali metals and alkaline-earth metals (except beryllium) form basic hydroxides in water, the alkaline-earth metals are in Group 2 rather than Group 1.

Educational objective:

Group 2 of the periodic table contains the alkaline-earth metals, which are a highly reactive family of metals with only two valence electrons that are readily lost to form cations with an oxidation state of +2. Alkaline-earth metals react with water to form hydrogen gas and either a basic hydroxide or an oxide, and the reactivity increases with increasing atomic number.